

Which object does the method change?

CONTEXT

prompt · examples · retrieved text

fixed weights
longer sequence

TRAINABLE STATE

soft prompt · adapter · LoRA

restricted writes
backprop through base

REPRESENTATION

integer codes · scales · metadata

approximate values
kernel-dependent speed



QLoRA composes these two

SEPARATE COST LEDGER — NOT COLUMN-MATCHED

BACKBONE STORAGE

shared or copied per task

INCREMENTAL STATE

prompt or learned delta

TRANSIENT WORK

tokens · activations · caches